

Jonathan Quinn

Zurich, Switzerland | Swiss, Canadian, German | jonfquinn@proton.me | +41 78 690 78 62 | github.com/jofras | jofras.github.io

EDUCATION

ETH Zurich

BSc in Computer Science

Zurich, Switzerland | Aug 2022 – Aug 2025

GPA: 5.2 / 6.0

- Thesis on ASR hallucination detection and embedding analysis (Supervised by Prof. Ryan Cotterell & Dr. Alexander Hoyle)
- Coursework: Machine Learning, Probability & Statistics, Numerical Methods, Deep Learning, Information Retrieval

Purdue University

Exchange Program, College of Science

West Lafayette, IN, USA | Aug 2024 – Dec 2024

GPA: 3.8 / 4.0

- Built 3D graphics pipeline from scratch in C++: vector operations and matrix transformations to rasterization, lighting models, texture mapping, and environment reflection using cubemaps, implementing real-time rendering with OpenGL shaders
- Extended Xinu OS with priority scheduling, memory management, and complete file system implementation in C

WORK EXPERIENCE

Infrastructure Developer

Service informatique de l'Entité neuchâteloise (SIEN)

Neuchâtel, Switzerland | Sep 2025 – Present

- Architected and implemented a complete overhaul of the in-house network scanner, achieving a 33x performance improvement (3hr to 5.5min runtime) and enabling hourly instead of daily scans for 1,800+ network devices
- Evaluated a dual-tunnel (device & user) Windows Always On VPN (AOVPN) proof of concept on FortiGate firewalls, configuring XML profiles, IKEv2, RADIUS/EAP authentication, and DHCP integration
- Built a full-stack network diagnostics application from scratch (Python backend, PHP/Laravel API, MySQL database, HTML/CSS frontend) used across the canton (50+ users), integrating team's requirements and enabling engineers to diagnose customer issues through automated path and port lookups, reducing configuration and troubleshooting time from minutes to seconds

Freelance Web Developer

3Steps Athletics

Remote | June 2025 – Present

- Designed and implemented bilingual business website for a sports organization using Next.js and Tailwind CSS for browsing and registration, deployed for spring 2026 camps

RESEARCH EXPERIENCE

Lost in Transcription: Embeddings from ASR Transcripts

Rycolab (ETH Zurich)

Bachelor's Thesis | Feb 2025 – Aug 2025

- Developed database-driven methodology for identifying corrupted sequences in ASR transcripts, processing 169M sentences using hash-based fingerprinting and run-length analysis
- Discovered and filtered 104K hallucinated sequences affecting 2% of podcast episodes, improving analysis' reliability
- Architected end-to-end NLP research pipeline for 50GB+ podcast corpus, including custom parallelized preprocessing, PostgreSQL-based quality control, and distributed training of Word2Vec/GloVe embeddings on HPC cluster
- Demonstrated systematic semantic divergence between podcast and news corpora through analysis of 2.3B tokens

SKILLS, AWARDS & ACTIVITIES

Programming

Python (scikit-learn, PyTorch, SpaCy, NLTK, NumPy, Pandas), SQL, Java, C/C++

Tools, Systems & Frameworks

Git/GitHub, Linux, Docker, Laravel/PHP, React/Next.js, SLURM

Languages

English (native), German (native), Swiss German (native), French (proficient)

Awards & Activities

- 2nd place, ETH x Optiver Hackathon (50 teams, 2024)
- Winner, Michigan State University Triathlon (2024)
- Ironman Finisher (2025)
- Laureate, Annual Highschool Literature Showcase
- Captain, GCK Lions Hockey Team (2016–2021)

Activities

Triathlon, Trail Running, Ice Hockey, Reading, Guitar, Travel